

Project Workflow

The project was executed through five main phases to transform raw transportation data into actionable business insights and strategic decision-making tools.

1. Data Modeling (ERD & Schema Design)

Objective

Design a structured and scalable database architecture.

Implementation

- Identified core entities such as:
 - Drivers
 - Trucks
 - Loads
 - Customers
 - Maintenance
- Defined:
 - Primary Keys (PK)
 - Foreign Keys (FK)
- Established relationships between tables to ensure:
 - Data integrity
 - Reduced redundancy
 - Efficient querying

2. Data Cleaning & Preparation (Python)

Objective

Improve data quality and ensure accurate analytical results.

Implementation

Using Python and the Pandas library, the following tasks were performed:

- Removed extra spaces and inconsistent formatting (Trimming)
- Handled missing values based on business rules
- Removed duplicate records

- Converted data types (especially dates and numeric fields)
- Created calculated columns such as:
 - Service Duration
 - Revenue Metrics

3. Database Implementation (SQL)

Objective

Store and manage cleaned data in a structured relational database.

Implementation

- Converted the ERD into physical database tables using SQL `CREATE TABLE` & `Alter table` statements
- Imported cleaned CSV datasets into the database
- Created SQL queries and joins to:
 - Connect related tables
 - Generate KPIs
 - Support reporting and analytics

4. Advanced Analytics & Customer Segmentation

Objective

Extract business insights and evaluate operational performance.

Implementation

- Classified customers into categories:
 - Platinum
 - Gold
 - Silver
- Calculated:
 - Revenue per Pound
 - Profitability metrics
- Analyzed customer account status:
 - Active vs inactive customers
- Evaluated operational and financial performance indicators

5. Data Visualization & Dashboard Development (Power BI)

Objective

Transform analytical results into interactive visual insights.

Implementation

- Connected Power BI to the SQL database
- Designed interactive dashboards displaying:
 - Total Revenue
 - Shipment Distribution Map
 - Fleet & Driver Performance
 - Preventive Maintenance Alerts
 - Customer Segmentation Insights

Technologies Used

- Python (Pandas, NumPy)
- SQL Server / MySQL
- Power BI
- CSV Data Sources

Expected Outcomes

- Improved data organization and accessibility
- Faster operational reporting
- Better customer segmentation
- Enhanced decision-making using interactive dashboards
- Early detection of fleet maintenance needs